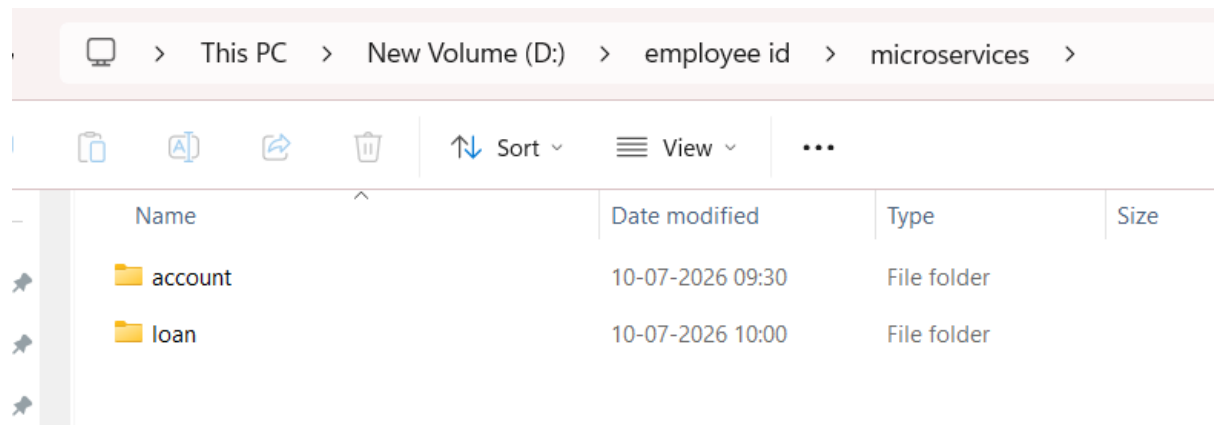
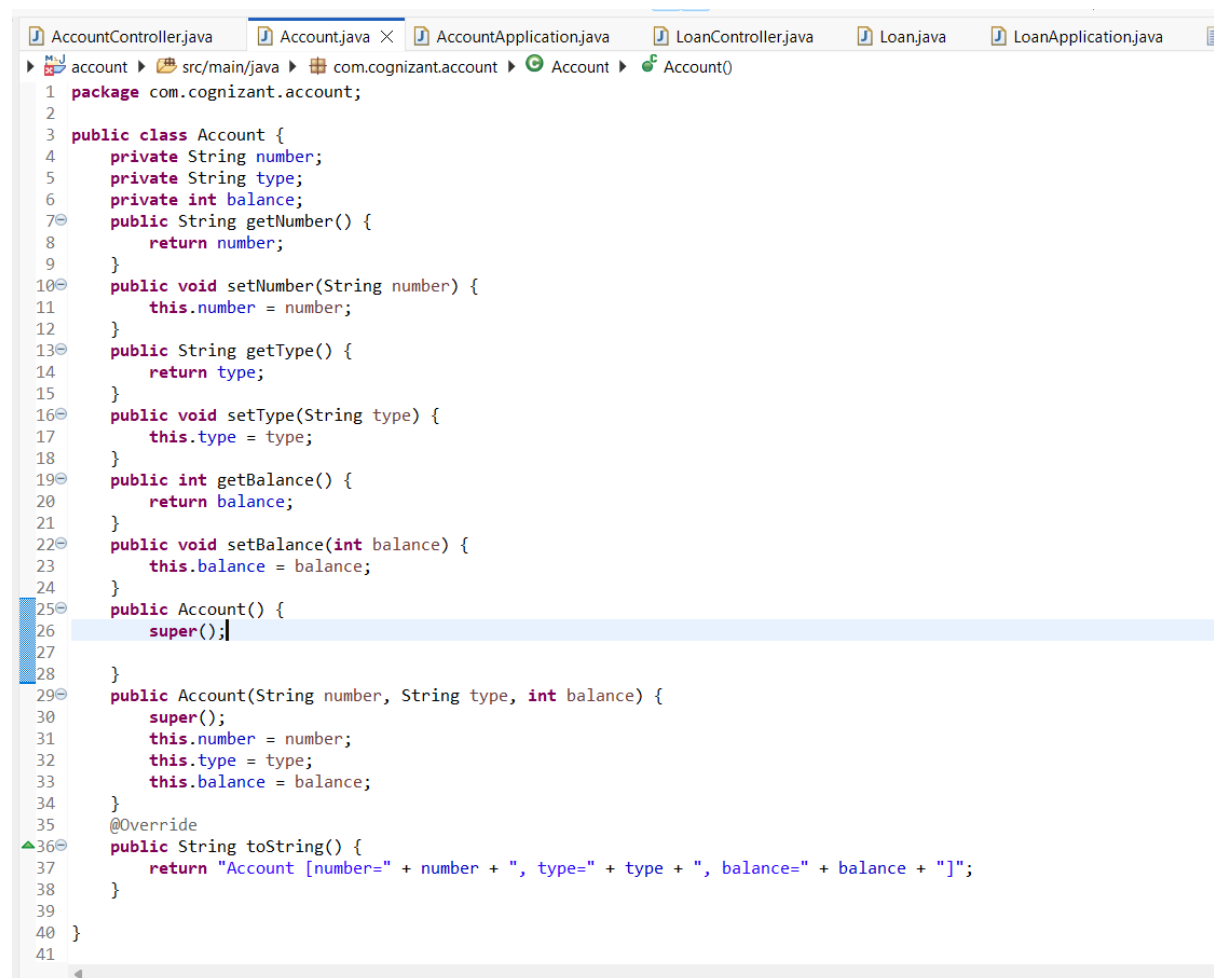


Exercise -1 Creating Microservices for account and loan



Account Service



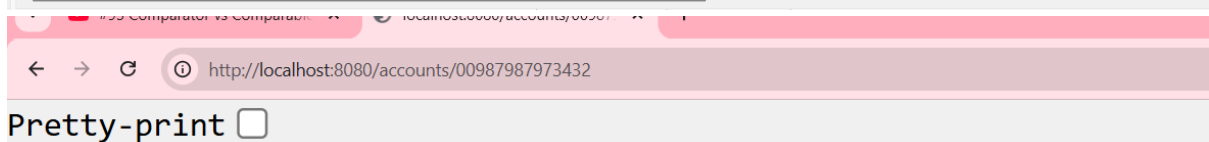
```
AccountController.java X AccountApplication.java LoanController.java Loan.j
account ▶ src/main/java ▶ com.cognizant.account ▶ AccountController ▶
1 package com.cognizant.account;
2
3 import org.springframework.web.bind.annotation.*;
4
5 @RestController
6 public class AccountController {
7
8     @GetMapping("/accounts/{number}")
9     public Account getAccount(@PathVariable String number) {
10         return new Account( number,"savings",234343);
11     }
12 }
13
```

```
AccountApplication.java X LoanController.java Loan.java LoanApplica
account ▶ src/main/java ▶ com.cognizant.account ▶ AccountApplication ▶
1 package com.cognizant.account;
2
3 import org.springframework.boot.SpringApplication;
4
5
6 @SpringBootApplication
7 public class AccountApplication {
8
9     public static void main(String[] args) {
10         SpringApplication.run(AccountApplication.class, args);
11     }
12 }
13
14
```

```
Console X
AccountApplication [Java Application] C:\Users\hemap\p2\pool\plugins\org.eclipse.justi.openjdk hotspot.jre.full.win32.x86_64_25.0.1.v20251108-1451\jre\bin\javaw.exe (10 Jul 2026, 10:10:22 am elapsed: 0:00:06) [pid: 32136]

:: Spring Boot :: (v4.1.0)

2026-07-10T10:10:23.409+05:30 INFO 32136 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Starting AccountApplication using Java 25.0.1 with P
2026-07-10T10:10:23.413+05:30 INFO 32136 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : No active profile set, falling back to 1 default pro
2026-07-10T10:10:23.501+05:30 INFO 32136 --- [account] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devto
2026-07-10T10:10:23.501+05:30 INFO 32136 --- [account] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting
2026-07-10T10:10:24.611+05:30 INFO 32136 --- [account] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-10T10:10:24.632+05:30 INFO 32136 --- [account] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-10T10:10:24.632+05:30 INFO 32136 --- [account] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-10T10:10:24.677+05:30 INFO 32136 --- [account] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed
2026-07-10T10:10:25.072+05:30 INFO 32136 --- [account] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path
2026-07-10T10:10:25.079+05:30 INFO 32136 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Started AccountApplication in 2.116 seconds (process
```



```
{"number": "00987987973432", "type": "savings", "balance": 234343}
```

Loan Service

```
AccountApplication.java  LoanController.java  Loan.java  LoanApplication.java
loan ▶ src/main/java ▶ com.cognizant.loan ▶ Loan ▶ Loan()
11 }
12 public void setNumber(String number) {
13     this.number = number;
14 }
15 public String getType() {
16     return type;
17 }
18 public void setType(String type) {
19     this.type = type;
20 }
21 public int getLoan() {
22     return loan;
23 }
24 public void setLoan(int loan) {
25     this.loan = loan;
26 }
27 public int getEmi() {
28     return emi;
29 }
30 public void setEmi(int emi) {
31     this.emi = emi;
32 }
33 public int getTenure() {
34     return tenure;
35 }
36 public void setTenure(int tenure) {
37     this.tenure = tenure;
38 }
39 public Loan(String number, String type, int loan, int emi, int tenure) {
40     super();
41     this.number = number;
42     this.type = type;
43     this.loan = loan;
44     this.emi = emi;
45     this.tenure = tenure;
46 }
47 public Loan() {
48     super();
49 }
50

loan ▶ src/main/java ▶ com.cognizant.loan ▶ LoanController ▶
1 package com.cognizant.loan;
2
3
4 import org.springframework.web.bind.annotation.*;
5
6 @RestController
7 public class LoanController {
8
9     @GetMapping("/loan/{number}")
10    public Loan getLoan(@PathVariable String number) {
11        return new Loan(number, "car", 400000, 258, 18);
12    }
13
14 }
15
```

AccountApplication.java LoanController.java Loan.java LoanApplicat

loan ▶ src/main/java ▶ com.cognizant.loan ▶ LoanApplication ▶

```
1 package com.cognizant.loan;
2
3 import org.springframework.boot.SpringApplication;
4
5
6 @SpringBootApplication
7 public class LoanApplication {
8
9     public static void main(String[] args) {
10         SpringApplication.run(LoanApplication.class, args);
11     }
12
13 }
14
```

```
1 spring.application.name=loan
2 server.port=8081
```

Console ×

LoanApplication [Java Application] C:\Users\hemap\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_25.0.1.v20251108-1451\re\bin\javaw.exe (10 Jul 2026, 10:11:39 am elapsed: 0:00:17) [pid: 25368]

Spring Boot (v4.1.0)

```
2026-07-10T10:11:41.040+05:30 INFO 25368 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : Starting LoanApplication using Java 25.0.1 with PID 25368 (D:\employee id\micros
2026-07-10T10:11:41.047+05:30 INFO 25368 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-10T10:11:41.139+05:30 INFO 25368 --- [loan] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false'
2026-07-10T10:11:41.139+05:30 INFO 25368 --- [loan] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' prop
2026-07-10T10:11:42.347+05:30 INFO 25368 --- [loan] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8081 (http)
2026-07-10T10:11:42.367+05:30 INFO 25368 --- [loan] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-10T10:11:42.368+05:30 INFO 25368 --- [loan] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-10T10:11:42.417+05:30 INFO 25368 --- [loan] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1278 ms
2026-07-10T10:11:42.838+05:30 INFO 25368 --- [loan] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8081 (http) with context path '/'
2026-07-10T10:11:42.847+05:30 INFO 25368 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : Started LoanApplication in 2.309 seconds (process running for 2.831)
```

← → ↻ http://localhost:8081/loan/H00987987972342

Pretty-print ☐

```
{"number": "H00987987972342", "type": "car", "loan": 400000, "emi": 258, "tenure": 18}
```